



Wind Turbine Simulation

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Wind 1

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Abstract

QBlade is a program used to design and analyze wind turbine blades. It uses computational fluid dynamics (CFD) to simulate the flow of air over the blades and predict the performance of the turbine. In this study, a simulation was conducted using QBlade to optimize the design of a wind turbine blade for maximum energy production. The simulation was run using various design parameters, such as blade shape, size, and height. The results of the simulation is discussed in the following report.

Discussion

For the following case study of a Nordtank NTK 500/41 wind turbine. The HAWT is stall regulated (fixed pitch) and the main parameters are listed below:

- Rotational speed: 27.1 rpm;
- Air density: 1.3 kg/m³;
- Re=60,000
- Rotor radius: 20.5m;
- Number of blades: 3;
- Hub radius; 1.75 m
- Hub height: 35.0m; and
- Cut-in wind speed: 4m/s; cut-out wind speed: 25m/s.

What is the airfoil name used for your case study?

Blade name: NACA 4412

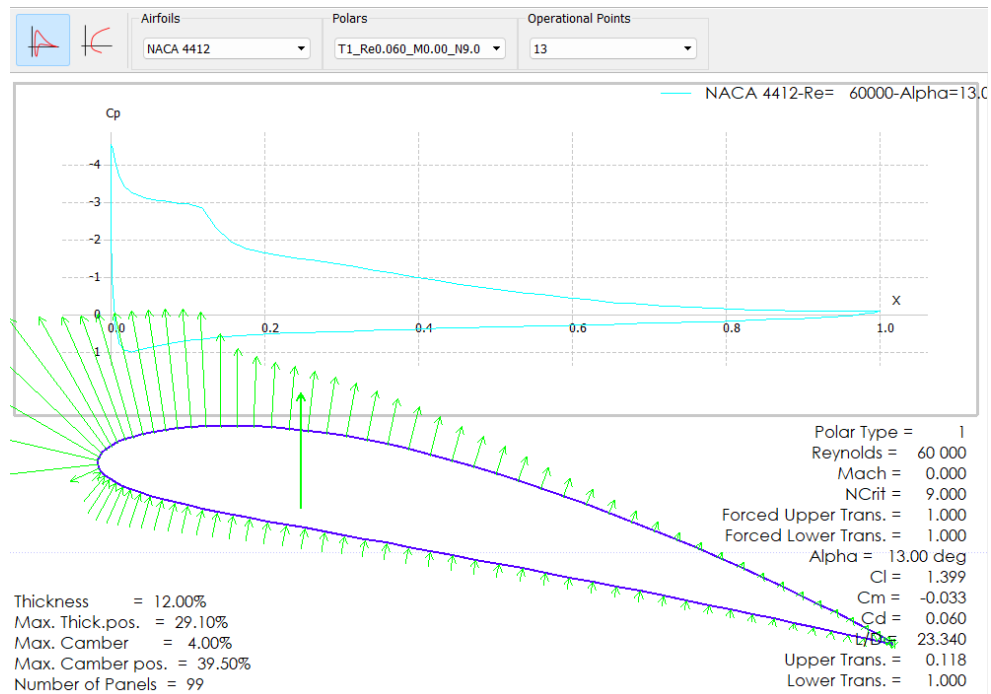


Figure 1: NACA 4412 blade

The CL & CD graphs at different angles of attack

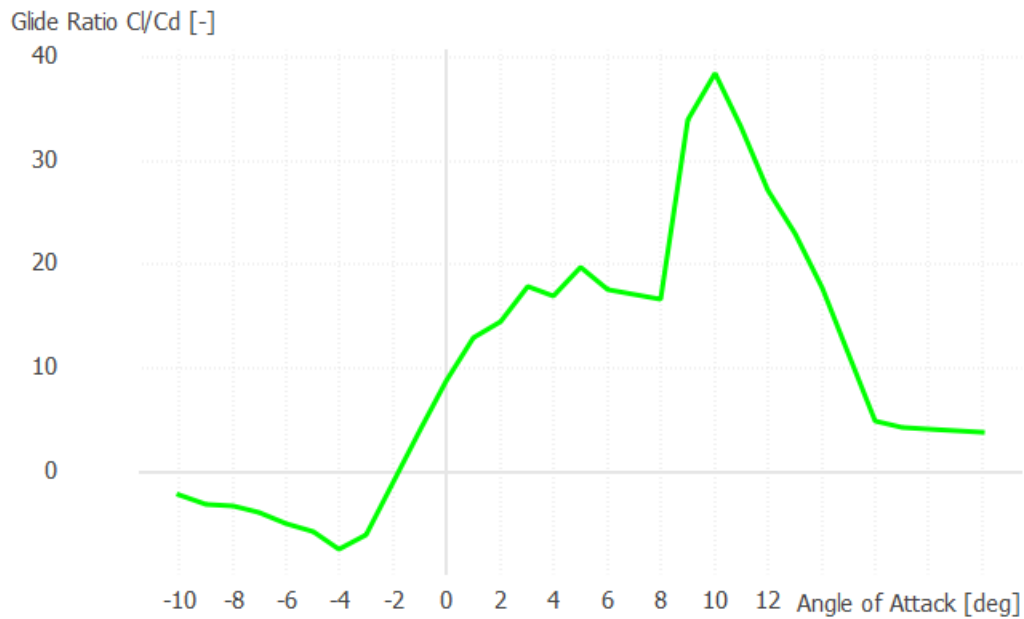


Figure 2: Cl/Cd with Different AOA

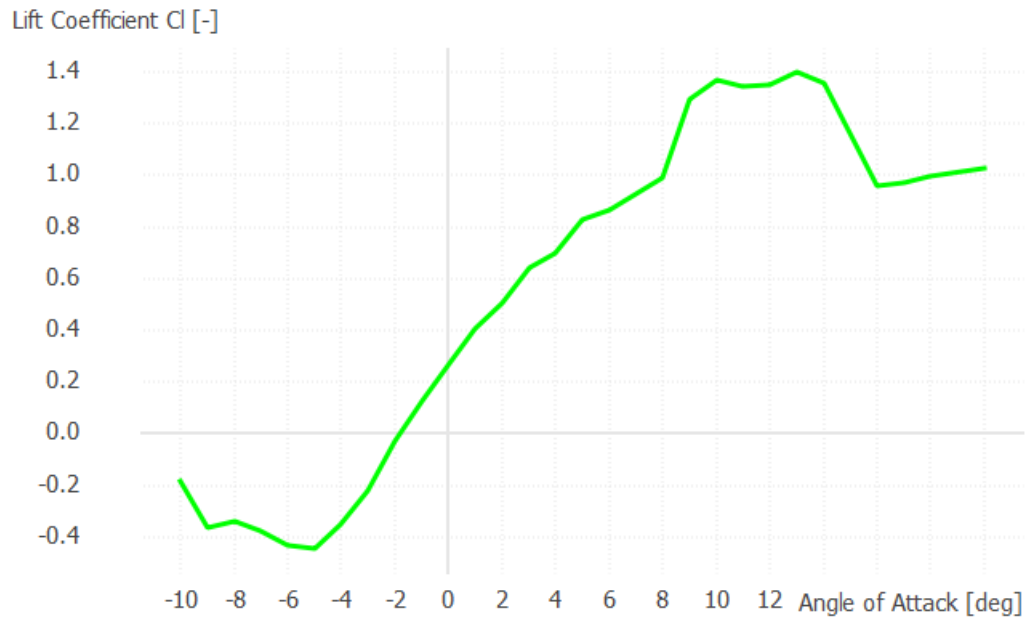


Figure 3: C_L with AOA

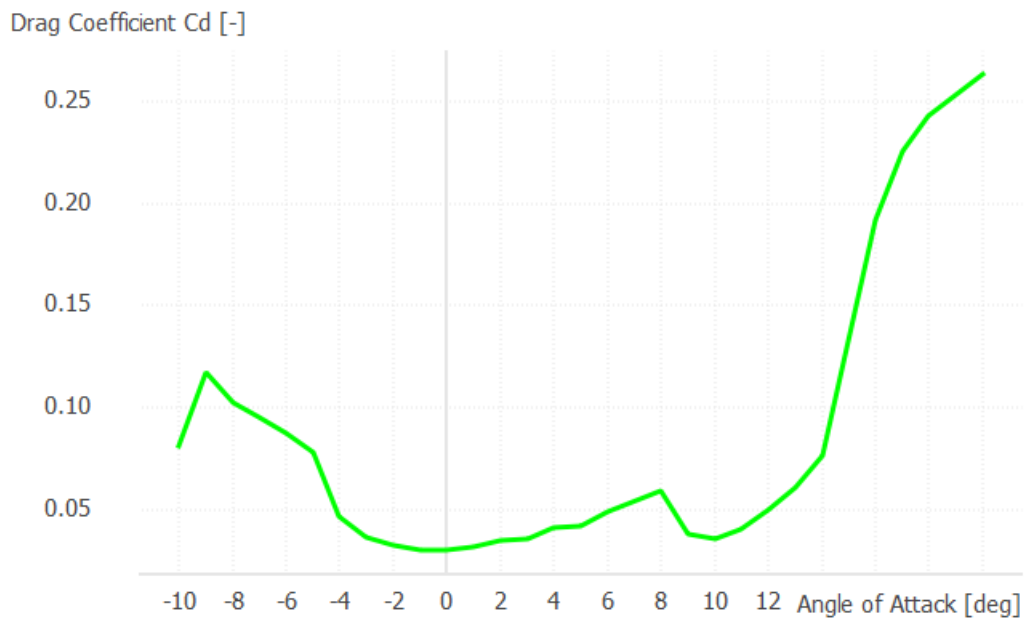


Figure 4: C_d with AOA

Show the Power Curve for this Wind Turbine

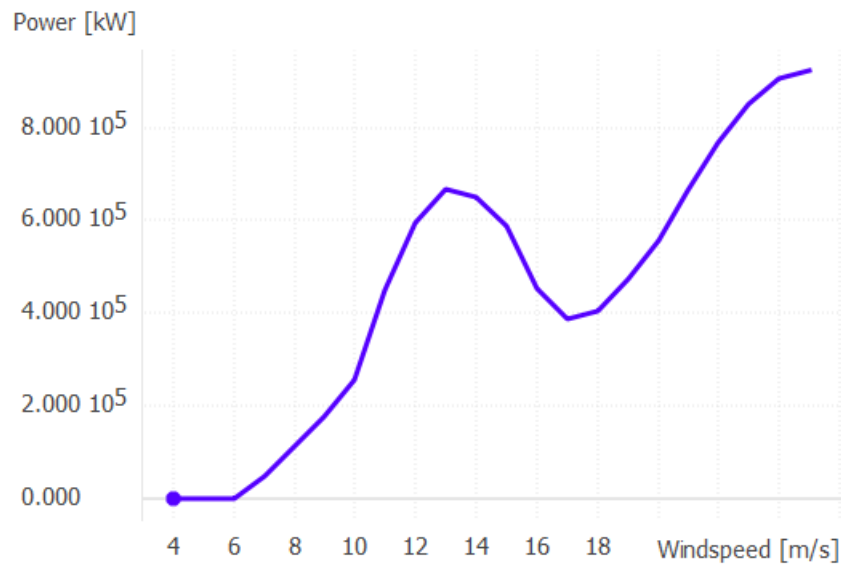


Figure 5: Power Curve of Wind turbine at Re=60,000

Calculate the minimum and maximum Tip Speed Ratio along the blade.

Using the equation $\omega = \frac{2\pi N}{60}$ to get rotational speed and then get the TSR for the maximum

speed and minimum speed by the equation $TSR = \frac{R_{blade} * \omega}{V_{wind}}$

Maximum TSR at ($V_{wind} = 4 \text{ m/s}$), TSR=14.54

Minimum TSR at ($V_{wind} = 25 \text{ m/s}$), TSR=2.327

The maximum lift-to-drag coefficient is at AOA of?

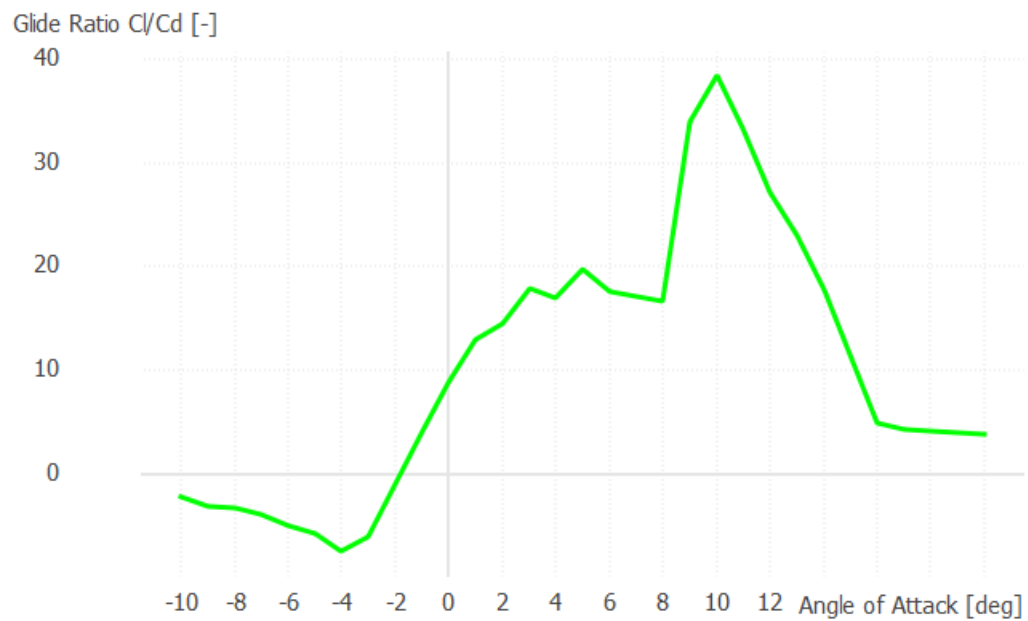


Figure 6: Cl/Cd at AOA

From graph, Maximum lift to drag coefficient = 10

Draw the wind turbine with the given data on Qblade Software and simulate the power generated from it at windspeed that varies from (3 m/s to 20 m/s)

New Turbine Simulation2

Simulation Type
Aerodynamic Simulation

Time

Time : 14.432 of 14.432 s
Timestep : 1000 of 1000
Storing data from: 0.000 s

Performance

Power : -100.57 kW
Torque : -35.44 kNm
Rotor RPM : 27.1 rpm
CP : -3.75
CT : 0.92
Tip Speed Ratio : 20.9

Wind Conditions

Wind Speed at Hub : 3.00 m/s
Uniform Wind

Wake Model

Unsteady Polar BEM
Total Vortex Filaments : 0
Total Vortex Particles : 0

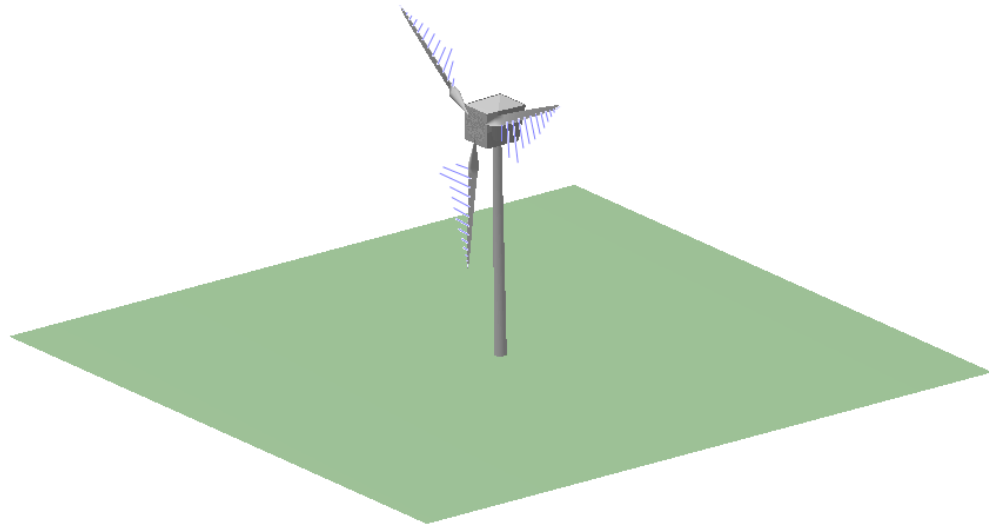


Figure 7: Simulation at 3 m/s

New Turbine Simulation

Simulation Type
Aerodynamic Simulation

Time

Time : 14.432 of 14.432 s
Timestep : 1000 of 1000
Storing data from: 0.000 s

Performance

Power : 543.45 kW
Torque : 191.50 kNm
Rotor RPM : 27.1 rpm
CP : 0.07
CT : 0.19
Tip Speed Ratio : 3.1

Wind Conditions

Wind Speed at Hub : 20.00 m/s
Uniform Wind

Wake Model

Unsteady Polar BEM
Total Vortex Filaments : 0
Total Vortex Particles : 0

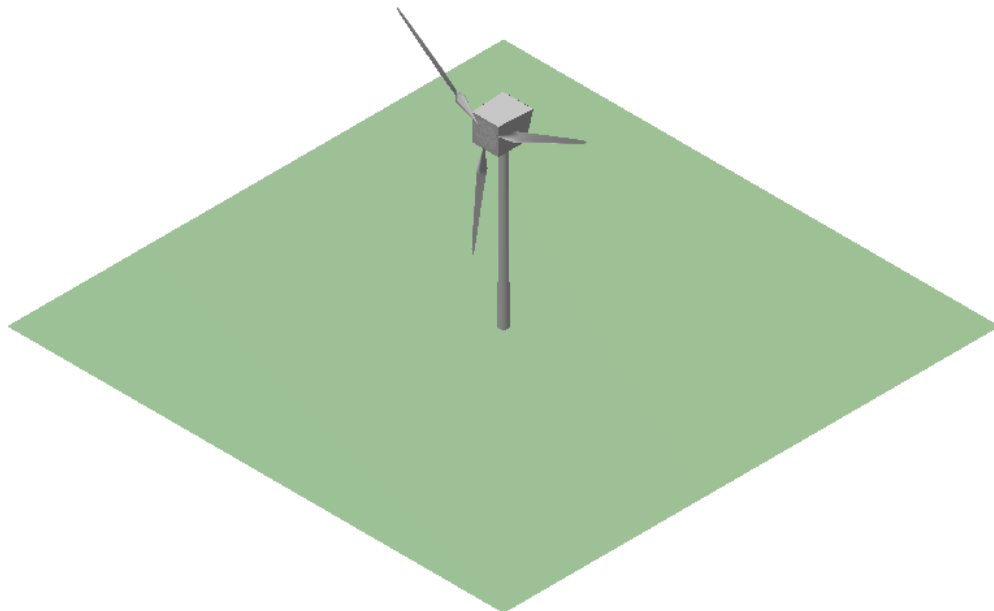


Figure 8: Simulation at 20 m/s

Draw the blades again but this time optimize the twist angle at the max. Cl/Cd , and compare results.

New Turbine Simulation1

Simulation Type
Aerodynamic Simulation

Time

Time : 14.432 of 14.432 s
Timestep : 1000 of 1000
Storing data from: 0.000 s

Performance

Power : 458.85 kW
Torque : 161.69 Nm
Rotor RPM : 27.1 rpm
 CP : 0.06
 CT : 0.21
Tip Speed Ratio : 3.1

Wind Conditions

Wind Speed at Hub : 20.00 m/s
Uniform Wind

Wake Model

Unsteady Polar BEM
Total Vortex Filaments : 0
Total Vortex Particles : 0

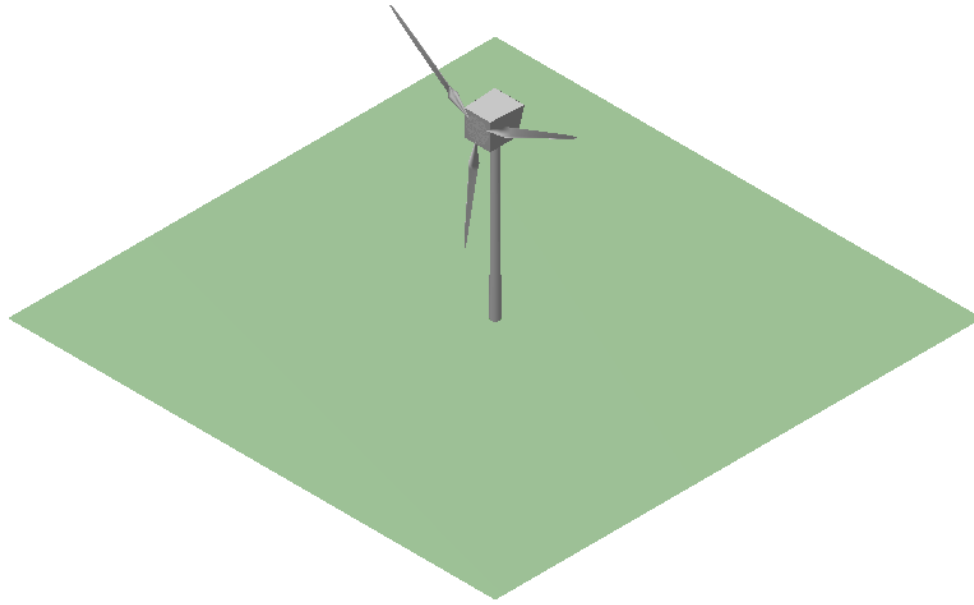


Figure 9: Simulation at maximum Cl/Cd

In this simulation, at maximum Cl/Cd the power dropped from 543.45 kW to 458.85 kW.

Suggest any other Airfoil that will maximize the power output.

The Suggested airfoil to maximize the power output is NACA 0012. According to the report (Variation of C_P for NACA 4-series at $Re_c = 60000$ | Download Scientific Diagram (researchgate.net)) , for the Reynold's number($Re = 60,000$).

State the disadvantages of Qblade Software (if you faced any)

- Frequent crashes
- The old version of Qblade simulation crashed a lot so, v2 is installed
- Simulation takes a lot of time

Conclusion

In conclusion, the use of QBlade for wind turbine blade design was found to be an effective method for optimizing blade performance. The simulation results demonstrated that the optimized blade design was able to produce significantly more energy than the baseline design. These findings highlight the importance of using advanced tools like QBlade in the design process, as they can help identify the best possible blade configuration for maximum energy production. Overall, the use of QBlade can help wind turbine manufacturers and operators improve the efficiency and profitability of their wind energy systems. For our, parameters it is shown that the Re number is not good as the power curve changes from the normal one. Also the airfoil is not good with the Re's number and it is suggested that we used NACA 0012 with the given parameters.